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PROCESSES FROM METHANE USING ASPEN HYSYS**

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**PONTIFÍCIA UNIVERSIDADE CATÓLICA DO RIO DE JANEIRO
CURSO DE ENGENHARIA QUÍMICA**

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PROJETO DE CONCLUSÃO DE CURSO

Projeto apresentado como requisito parcial para
obtenção do grau de Bacharel em Engenharia Química pelo
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SIMULATION OF METHANOL PRODUCTION PROCESSES FROM METHANE USING ASPEN HYSYS

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ABSTRACT

Global warming and the growing energy demand have created the necessity of developing alternate fuels and sustainable and clean production routes. Within the approaches to solve this issue, the commonly named methanol economy stands out, since methanol, as fuel, has advantages that go from the facility to storage and transport to its properties as an alternate fuel. However, the traditional process to produce it from natural gas (steam reforming), depends on large energy expenditure and capital cost. It also emits more carbon dioxide, contributing to global warming. In this context, the present work uses simulation with the Aspen Hysys v.8.8 software and its economic evaluation and energy analysis tool to demonstrate that an alternative route to produce methanol, from biogas, using carbon dioxide (dry reforming), is more advantageous both to the environment and to the investor as it demands less energy, produces more methanol and costs less to install and operate.

Keywords: methanol, biogas, simulation, Aspen Hysys, global warming, economic evaluation, sustainability.

1. Introduction

According to the Goddard Institute for Space Studies (GISS), earth's surface temperature has increased by about 0.8 °C since 1880 [1]. This trend will continue as humankind continues to harm the environment and scientists predict an increase of 2 to 6 °C on the average surface temperature by the end of the 21st century, as predicted by model simulations by the Intergovernmental Panel on Climate Change (IPCC) [2]. The best-case scenarios assume that greenhouse gas emissions will grow slowly while worst-case scenarios assume the society will burn more and more fossil fuel [1]. According to IPCC, it's likely that anthropogenic greenhouse gas increases caused most of the increase in global temperature, while the natural forcing factors would be likely to have produced cooling, not warming [2].

The humankind heavy dependence on fossil fuels caused a 30 % increase of carbon dioxide concentration in the atmosphere since 1950. In 2015, 45 % of the CO₂ emissions came from coal burning, 35 % from oil burning and 20 % from natural gas burning [3]. In Figure 1, is shown the increase in the concentration of carbon dioxide from 1980 to 2018, when the atmosphere CO₂ concentration reached levels near 410 ppm [4]. Therefore, the development of alternative fuels, more environmentally friendly, makes itself necessary to reduce or stop this increase in CO₂ emission. The hydrogen economy and the methanol economy are two of the main approaches to solve this problem.

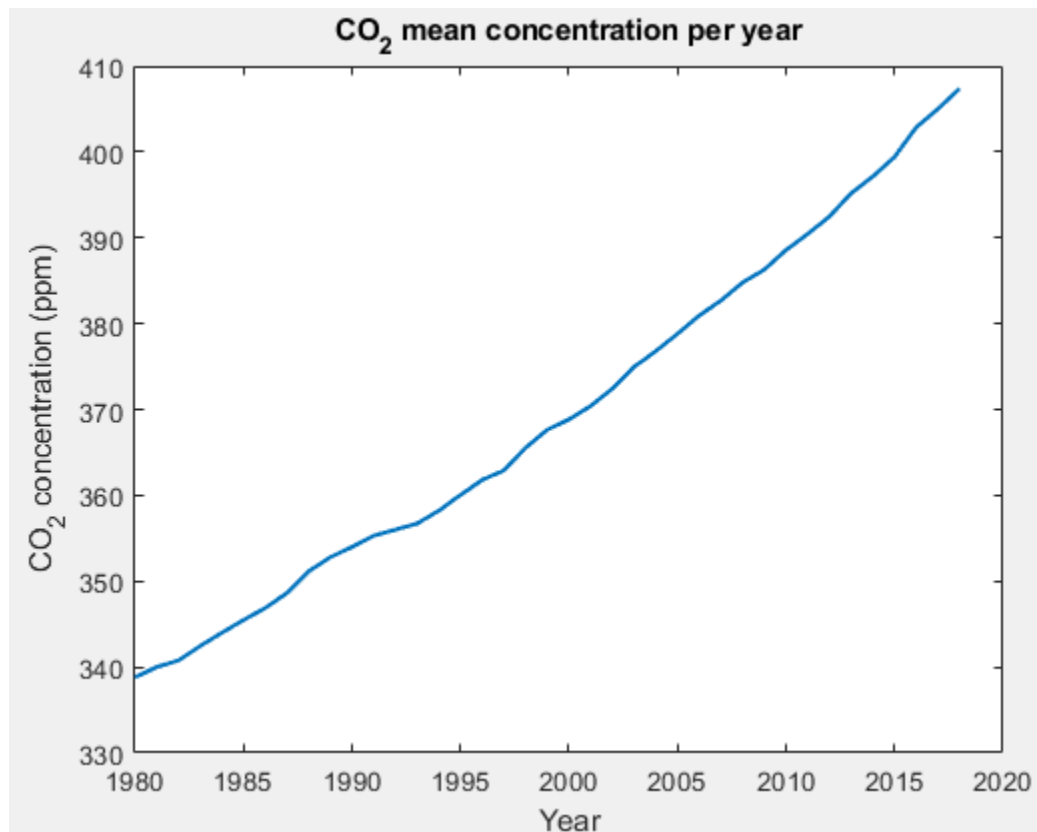


Figure 1. Concentration of carbon dioxide in the atmosphere [4]

Hydrogen is generated by water electrolysis and subsequently used as a clean fuel, generating only water in its combustion. Due to its extreme volatility and explosive nature, hydrogen has serious limitations in terms of storage, transportation and deliverance of energy, while methanol, an oxygenated liquid hydrocarbon, represents safe means of storage and transport and excellent fuel (when combined with dimethyl ether) [5].

Methanol is among the top five of most widely traded chemicals in the world, used to produce adhesives, paints, LCD screens, silicones, pharmaceuticals [6] and single-cell proteins [7]. It can also be used as energy storage and fuel, or as starting material for chemicals such as formaldehyde, acetic acid, ethylene and propylene, the two last used as feedstock for the production of synthetic hydrocarbons [5]. Its market is

dominated by few players, such as Methanex, SABIC, Methanol Holdings Ltd, LyondellBasell Industries Holdings BV and Petronas [8].

The performance enhancing attributes of methanol were realized early on and used in motor racing (especially blends containing benzene and methanol) and aviation, because of its benefits in terms of octane number and latent heat, but since this approach requires dual fuel systems, it was foreshadowed by single fuel approaches. In the 80's decade, in California, vehicles using methanol as fuel were tested (more than 15000 were sold and operated), causing significantly lower unburned hydrocarbon and oxides of nitrogen emissions than typical gasolines [6]. Methanol is produced mainly from coal, natural gas, biogas, coke-oven gas, hydrogen and biomass [9].

Natural gas is a clean and effective energy source since its combustion generates fewer greenhouse gases than coal or petroleum liquid [10]. Methane is the predominant component of natural gas and it is found plentifully in many locations, therefore most studies on its use focus on methane activation [11]. It is also the main component of biogas, which is generated as a waste product of microorganisms and also contains CO₂. The methanogenesis, biological process for biogas generation by anaerobic methanogenic bacteria, was discovered in 1776 by Alessandro Volta and has been used since the 19th century in anaerobic digestion reactors [5].

For methanol production, the most studied route uses two steps: syngas production from methane and CO₂ hydrogenation to methanol [5], with further separation as shown in Figure 2. The different routes to produce syngas from methane (biogas) are shown in Figure 3.

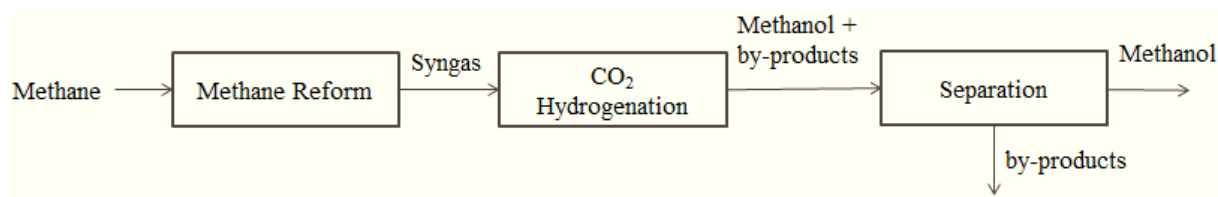


Figure 2. Block diagram of the multistep route of methanol synthesis

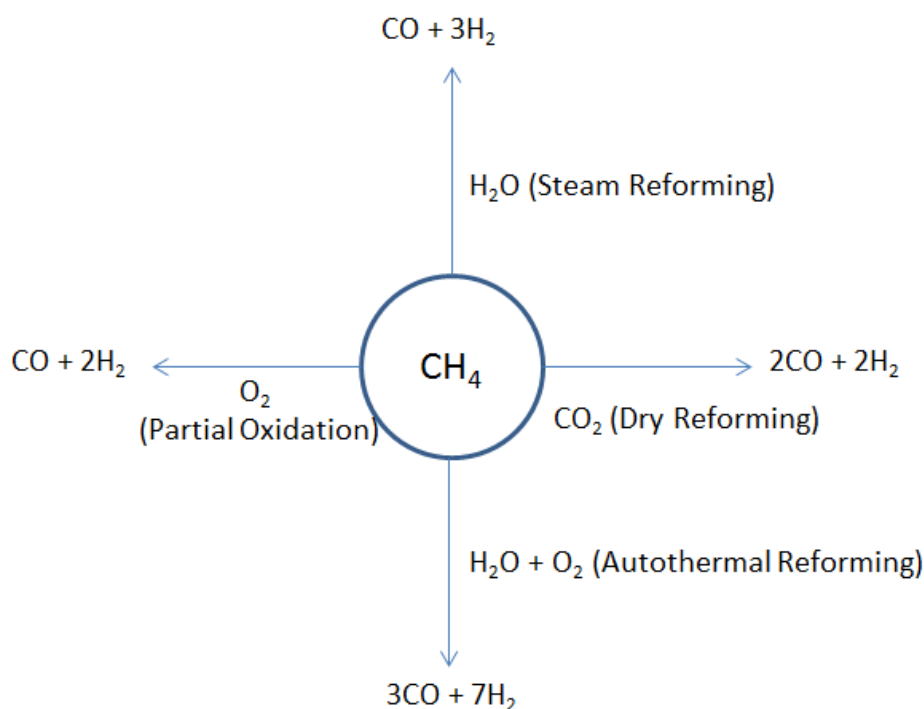


Figure 3. Different routes and reagents of multistep methanol synthesis

Syngas (synthesis gas), the most usual intermediary in multistep processes of methanol production from methane, is a combustible gas, consisting mostly of hydrogen and carbon monoxide, and can be used to produce electricity in steam cycles and fuel cells although it's not as efficient as other energy sources, being mostly used as feedstock to produce transportation fuels, such as methanol, DME (dimethyl ether) and diesel [13]. The syngas composition is characterized by the stoichiometric number S , which should, ideally, be equal or slightly above 2 for the production of methanol.

$$S = \frac{n_{H_2} - n_{CO_2}}{n_{CO} + n_{CO_2}} \quad (1)$$

where n_{H_2} , n_{CO_2} and n_{CO} are, in the following order, the number of moles of hydrogen, carbon dioxide gases and carbon monoxide [5].

The multistep synthesis of methanol from methane is usually done by using steam or oxygen (steam reforming, partial oxidation or autothermal reforming) [10]. However, the traditional processes are highly endothermic, using high temperatures and excessive energy which increases the cost. Moreover, as it will be shown later, a parallel reaction (water-gas shift) produces CO_2 , making the route environmentally unattractive. It also involves water evaporation, making it technologically more complex [14].

Due to those disadvantages of the traditional reforming processes and the increasing effects of global warming, shifting the focus to new, environment friendly and cost-effective processes is necessary, basing the research in economic, energetic and sustainability metrics. Since the traditional steam reforming route is more technologically complex and contribute to greenhouse gases emission, producing CO_2 , the dry reforming, which is a route that uses CO_2 to produce syngas, can become a great alternative [14]. The dry reforming process is highly endothermic ($\Delta H^0_{298K} = 70.7 \text{ kcal mol}^{-1}$) [5] while the pressure of the process should be low. Research on dry reforming is dominated by studies on active, stable and selective catalysts, especially nickel-containing materials, supported on basic oxides [14].

The use of environmental metrics such as material intensity (which measures material waste), energy intensity (which measures energetic waste) [15] and accumulation factor [16] can help evaluate the waste produced, help fulfill regulatory compliance and improve corporate reputation for investors and insurers [15]. In Table 1 are shown the equations for the aforementioned metrics.

Table 1. Definition of each sustainability metric used

Metric	Equation	Reference
Material Intensity (M)	$M = \frac{m_{product}}{m_{raw\ materials}}$	[15]
Energy Intensity (E)	$E = \frac{E_{consumed}}{m_{product}}$	[15]
Accumulation Factor (AF)	$AF = \frac{m_{recycle}}{m_{leaving\ recycle}}$	[16]

Based on what was presented, the objectives of this work are to simulate, using Aspen Hysys v.8.8 software, methanol production using steam, autothermal and dry reforming based processes, generating process diagrams for each route. It also aims to compare the economic and sustainability metrics related to each synthetic route, hoping to move towards substituting the environmentally unfriendly traditional multistep route via steam reforming with a more sustainable and economically advantageous process.

2. Process Description and Simulations

As it was mentioned, methanol can be synthesized, from methane, in two main routes: syngas-based multistep process or single-step processes such as oxidative coupling and direct oxidation (DMTM) [10]. The multistep processes via steam reforming, steam reforming followed by partial oxidation (autothermal reforming) and dry reforming were simulated in stationary state using the process simulator Aspen Hysys v8.8, using Peng-Robinson property package, since it's the most enhanced model of the software with the largest applicability range in terms of temperature and pressure [17] and the components are mostly nonpolar liquid or gas mixtures.

2.1.Steam Reforming

In the most traditional route, methane reacts with steam at high temperatures to form CO and H₂, with a stoichiometric number *S* approximately 3. Part of the CO

consequently reacts with the steam (water-gas shift reaction) to yield more H₂ and also CO₂ [5]. Both reactions and respective enthalpies are shown in Table 2. The pressure inside the reactor usually sits between 3 to 25 atm depending on the technology used [18]. According to Le Chatelier's principle [19], lower pressures result in higher reaction conversions.

Table 2. Reactions involved in steam reforming of methane

Reaction	ΔH_{298K} (kcal mol ⁻¹)	Reference
$\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$	49.1	[5]
$\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	-9.8	[20]

Traditionally, nickel is the most studied metal for steam reforming because it's cheap and fairly active [21]. The nickel surface is directly related to the activity of the catalyst which, for low temperature service, contain nickel crystals below 5 nm and dispersion of 10 % or higher [18]. Ni supported on Al₂O₃ is promising due to its low price and significant activity, which increases with nickel content. However, the more nickel sites the more its aggregation, being 12 wt. % a typical nickel content to avoid this aggregation [14].

Carbon also has been studied as catalysts (activated carbon) due to its porous structure, tunable surface chemistry and low cost [22]. It's also possible to use carbon as supports because of the lower reaction temperature (typically between 200 and 400 °C), although it's used mostly in methanol steam reforming. Aluminum has also been shown to increase the catalyst activity, especially when combined with magnesium in conventional Ni/Al catalysts [21].

At industrial processes, the water-gas shift reactions are carried out in two steps: a high temperature shift (HTS) at 350 to 450 °C and a low temperature shift (LTS) at

180 to 250 °C. [23] This reaction helps to avoid deactivation of the catalyst (usually Ni) due to CO deposition. It also reduces CO concentration and produces additional H₂. It is an overall exothermic process and the catalysts have a few common characteristics including available oxygen vacancies, activity in dissociation of water and low CO adsorption strength [21]. It's a reversible reaction and the thermal equilibrium is reached faster at higher temperatures, but low temperature catalysts (LTC) provide high conversion although they are kinetically limited, which is attributed to the balance between Le Chatelier's principle and Arrhenius law [24].

2.2. Autothermal Reforming

The autothermal reforming process, which basically consists in adding oxygen to the syngas produced by the steam reforming process [25] helps reducing the stoichiometric number *S* to approximately 2, the ideal condition to produce methanol with the multistep route [5]. The reactions involved in the process are shown in Table 3.

Table 3. Reactions involved in autothermal reforming of methane

Reaction	ΔH_{298K} (kcal mol ⁻¹)	Reference
$\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$	49.1	[5]
$\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	-9.8	[20]
$\text{CH}_4 + \text{O}_2 \rightarrow \text{CO} + 2\text{H}_2$	-8.60	[26]

Autothermal reforming can use steam and oxygen in a two-step process consisting of steam reforming followed by partial oxidation, but there is also stand-alone autothermal reforming technology which merges partial combustion and catalytic steam reforming in one compact refractory lined reactor [10].

Combining an exothermic reaction and an endothermic reaction can improve not only the energy efficiency but also the control of the reactor's temperature, avoiding catalyst deactivation by sintering [27]. The autothermal reforming can have two steps,

with a primary reformer producing excess of H₂ and a secondary reformer producing methanol from this excess, creating an energy efficient production of syngas, or one step using stand-alone autothermal reforming technology, yielding syngas for methanol production in large scale [10].

2.3. Dry Reforming

The dry reforming process is based on the methane reaction with CO₂, obtaining syngas with 1:1 H₂ to CO ratio. The reaction and its enthalpy are shown in Table 4. While this low ratio is not considered ideal for methanol synthesis [5], the consumption of CO₂ and the low temperatures, combined with the lower operational cost for using only one reactor, makes this route worthy of deeper study.

Table 4. Dry reforming of methane reaction

Reaction	ΔH_{298K} (kcal mol ⁻¹)	Reference
$\text{CH}_4 + \text{CO}_2 \rightarrow 2\text{CO} + 2\text{H}_2$	59.1	[5]

While the thermodynamics for this process are not as favorable as the autothermal reforming or the steam reforming, the consumption of one mole of carbon dioxide per mole of methane invested could reduce the carbon impact, which could overall lead to a “greener” consumption of methane [28].

The major obstacle preventing the commercialization of this process is that carbon formation and sintering quickly deactivates conventional reforming catalysts. Noble metal-based, partly sulfur passivated Ni and noble metal-promoted Ni catalysts have good potential for successful application of full-scale reforming plants. However, small feeds of steam are needed as co-feed to eliminate severe carbon formation [29].

Catalysts for this process suffer from deactivation caused by catalytic coke which rate is influenced by temperature. Higher rate of coking is promoted by lower

temperatures, below 700 °C [30]. Noble metal catalysts show high activity and good stability against coking while nickel catalysts are not resistant against coking [28].

The dry reforming is also considered suitable for other processes such as iron ore reduction and Fischer-Tropsch synthesis. Olah et al. (2011) consider the dry reforming suitable for methanol synthesis if hydrogen generated from other sources is added to the syngas produced [5]. However, it will be shown further ahead that such addition is probably unnecessary.

2.4. CO₂ Hydrogenation

The methanol production through the syngas routes can be described by three equations, as shown in Table 5. This kinetic mechanism still isn't consensual, thus the reactions are dependent and the process can be described, for simplicity, by any combination of two reactions [31]. Since methanol synthesis is exothermic and involves a decreasing number of gaseous moles, low temperatures and high pressures are more adequate to reach better conversion [10]. At the present work, it will be assumed the occurrence of CO₂ hydrogenation and water-gas shift reactions.

Table 5. Reactions involved in methanol synthesis from syngas

Reaction	ΔH_{298K} (kJ mol ⁻¹)	Reference
CO₂ + 3H₂ → CH₃OH + H₂O	-40.9	[10]
CO + 2H₂ → CH₃OH	-90.7	[10]
CO₂ + H₂ → CO + H₂O	49.8	[10]

3. Results and Discussion

The steam reforming, autothermal reforming (bi-reforming) and dry reforming processes described beforehand, were simulated using Aspen Hysys v8.8. Reactors based on equilibrium reactor model were used for all the processes, since it doesn't

demand kinetics constant values. In total, were obtained 3 process diagrams and reports for comparison of the synthetic routes. The tags for each vessel, column or heat exchanger are given according to Table 6.

Table 6. Tags for each model simulated in the processes, where ABC indicates the number of the model in order of appearance

Model	Tag
Equilibrium Reactor	ERV-ABC
Heat Exchanger	E-ABC
Splitter	X-ABC
Distillation Column	T-ABC
Flash Separator	V-ABC
Recycle Tool	RCY-A

3.1. Steam Reforming

The steam reforming step uses 3 different reactors: the methane reformer (ERV-100), the high temperature shift reactor (ERV-101) and the low temperature shift reactor (ERV-102), and all of them use equilibrium reactor models. The water-gas shift reactors work 450 °C and 180.47 °C, respectively, and 100 kPa. The temperature of the methane reformer was set at 826.85 °C using Aspen Hysys' optimization tool.

The optimization tool was also used to set the methanol synthesis reactor (ERV-103) temperature, at 12.23 °C. The top stream from this reactor (ERV-103), containing mostly hydrogen and methanol, goes through a flash separator (V-100) where the hydrogen is separated and reinstated to the process as a recycle stream. The complete process diagram is shown in Figure 4. The bottom stream from the separator goes through a distillation column (T-100) containing 20 trays, resulting in a final product stream containing 99.99 % of methanol. In Tables 7 and 8 are shown the process conditions, molar flow and composition of the main streams.

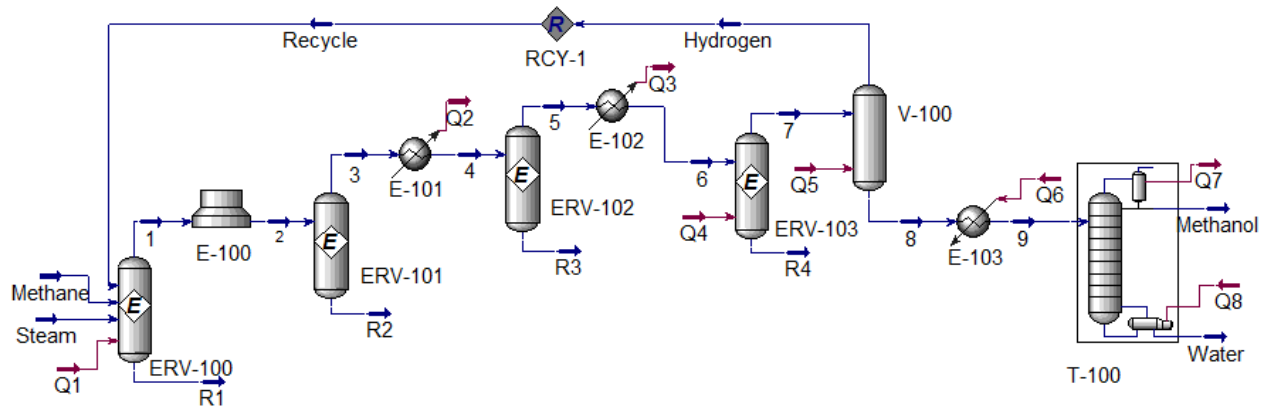


Figure 4. Process diagram of multistep methanol synthesis using methane steam reforming

Table 7. Process conditions and molar flow of the main streams in the steam reforming process

	Methane	Steam	Methanol
Temperature (°C)	25	100	56.69
Pressure (kPa)	100	100	100
Molar Flow (kgmol h⁻¹)	100	100	94.66

Table 8. Molar composition of the main streams in the steam reforming process

Component	Molar Composition		
Stream	Methane	Steam	Methanol
Methane	1	0	$4.70 \cdot 10^{-5}$
Methanol	0	0	0.9999
CO	0	0	$2.90 \cdot 10^{-11}$
CO₂	0	0	$4.30 \cdot 10^{-7}$
H₂O	0	1	$1.14 \cdot 10^{-7}$
Hydrogen	0	0	$4.90 \cdot 10^{-5}$

3.2. Autothermal Reforming

The autothermal reforming process uses the same reactors as the steam reforming process, with the addition of a partial oxidation reactor (ERV-103). In this case the product stream of the methanol synthesis reactor contains mostly methanol, methane and oxygen. The latter are separated in the flash separator (V-100) and sent back to the process as a recycle stream, while the bottom stream goes through a distillation column (T-100) containing 20 trays to separate the water. The process diagram is shown in Figure 5. The process conditions, molar flows and composition of the main streams are shown in Tables 9 and 10.

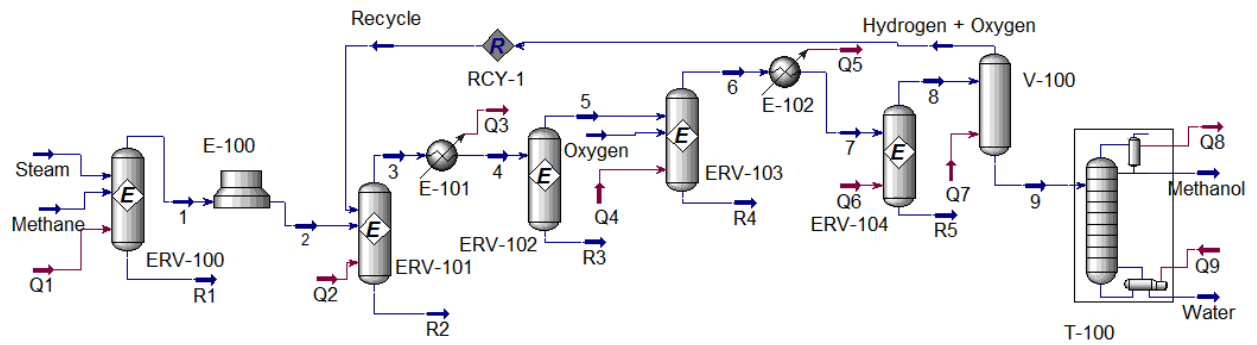


Figure 5. Process diagram of multistep methanol synthesis using methane steam reforming followed by partial oxidation

Table 9. Process conditions and molar flow of the main streams in the autothermal reforming process

	Methane	Steam	Oxygen	Methanol
Temperature (°C)	25	100	200	46.53
Pressure (kPa)	100	100	100	100
Molar Flow (kgmol h ⁻¹)	100	100	120	98.66

Table 10. Molar composition of the main streams in the autothermal reforming process

Component	Molar Composition			
Stream	Methane	Steam	Oxygen	Methanol
Methane	1	0	0	$1.93 \cdot 10^{-18}$
Methanol	0	0	0	0.9997
CO	0	0	0	$1.00 \cdot 10^{-12}$
CO ₂	0	0	0	$1.40 \cdot 10^{-07}$
H ₂ O	0	1	0	$3.78 \cdot 10^{-08}$
Hydrogen	0	0	0	$2.23 \cdot 10^{-05}$
Oxygen	0	0	1	0.0003

3.3. Dry Reforming

The dry reforming process, which process diagram is shown in Figure 6, uses 2 different reactors: the methane reformer (ERV-100) and the methanol synthesis reactor (ERV-101). The molar flows of the feed streams (Methanol and CO₂) were set at 100 kgmol h⁻¹ (1:1). The dry reforming reaction occurs on the ERV-100 reactor (equilibrium reactor model) at 900 °C [26] and 100 kPa and the product is cooled to 15 °C to enter the methanol synthesis reactor (ERV – 101), which operates at 7.5 °C. The top product goes through a splitter (X-100) which simulates a PSA column that separates the carbon monoxide and a flash separator (V-100) which bottom product (mostly methanol) is mixed with the bottom product from ERV-101 (also mostly methanol) to increase the global conversion of the process. The molar flow of the methane stream and the carbon dioxide stream were set on 100 kgmol h⁻¹.

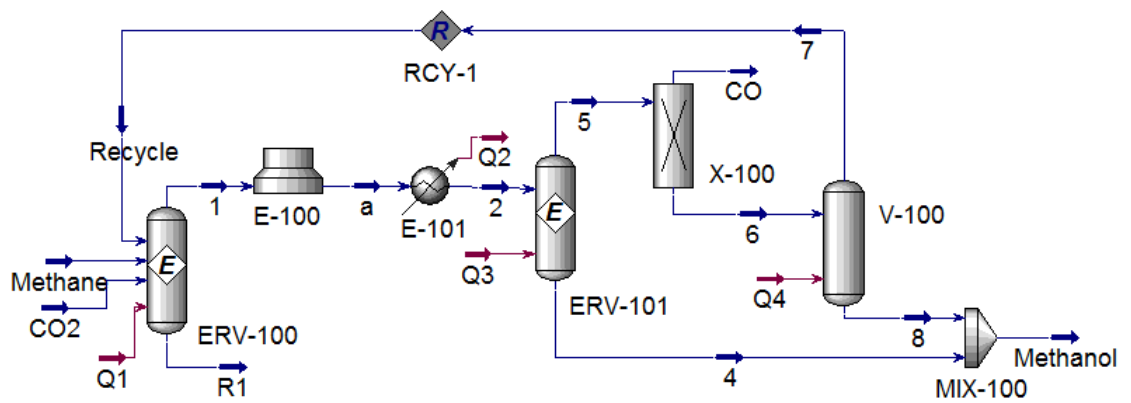


Figure 6. Process diagram of multistep methanol synthesis using methane dry reforming

In Tables 11 and 12 are shown temperature, pressure, molar flow and composition of the 2 feed streams and the final product stream. The temperature of the CO_2 hydrogenation was set using the case studies tool, which demonstrates that the temperature of the reactor influences both the fraction and the flow of methanol in the product stream. As the temperature increases, the fraction of methanol increases and the flow of methanol decreases, as shown in Figures 7 and 8. The temperature was set to obtain the better conversion possible with a product stream containing 99.95 % or more methanol.

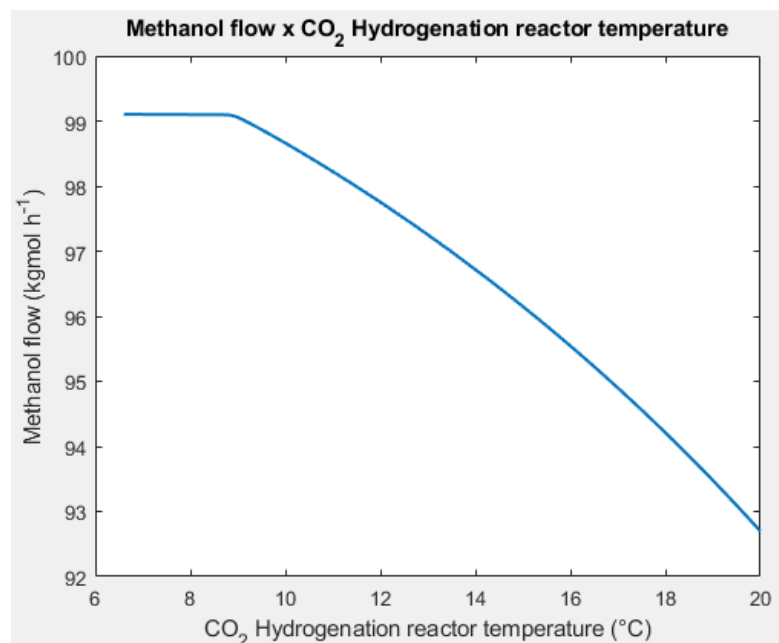


Figure 7. Case studies of methanol molar flow in the final product stream at different temperatures of the methanol synthesis reactor (ERV-101)

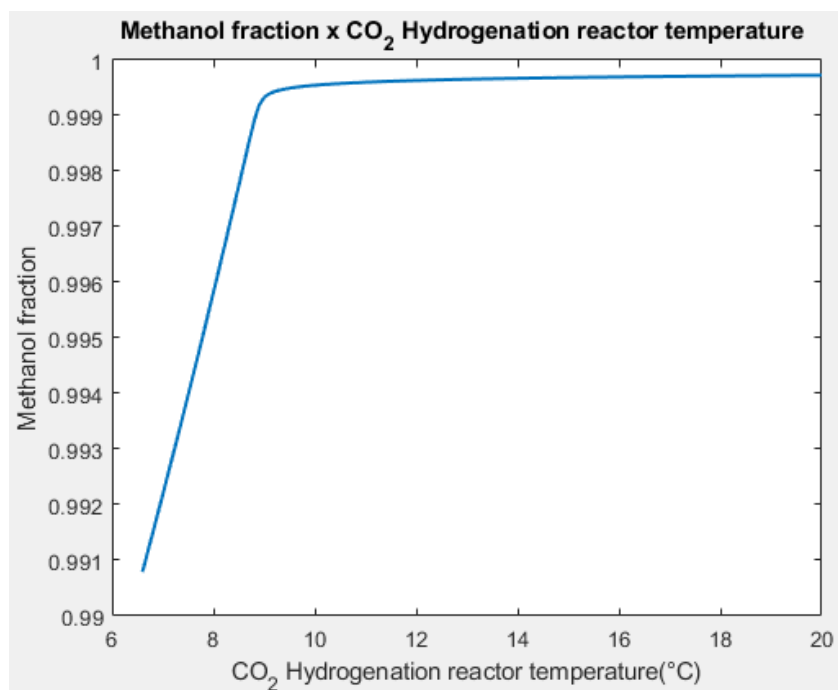


Figure 8. Case studies of methanol fraction in the final product stream at different temperatures of the methanol synthesis reactor (ERV-101)

Table 11. Process conditions and molar flow of the main streams in the dry reforming process

	Methane	CO ₂	Methanol
Temperature (°C)	25	100	3.8
Pressure (kPa)	100	100	100
Molar Flow (kgmol h ⁻¹)	100	100	100.03

Table 12. Molar composition of the main streams in the dry reforming process

Component	Molar Composition		
Stream	Methane	CO ₂	Methanol
Methane	1	0	0.0002
Methanol	0	0	0.9995
CO ₂	0	1	0.0002
CO	0	0	0.0001
H ₂ O	0	0	0
Hydrogen	0	0	0

Using the conditions described, the product stream flow is 100 kgmol h⁻¹, with 99.95 % of methanol and small concentration of methane, CO₂ and CO, which amounts to 99.95 kgmol h⁻¹ of methanol produced. With a similar process, without the recycle stream, results in 99.7 kgmol h⁻¹ of product stream and 99.40 % of methanol, amounting to 99.11 kgmol h⁻¹. Although there wasn't significant increase to the methanol production, the increase in the purity of the product stream shows the importance of the recycle stream.

The simulation results show an increase of 5.6 % of the methanol produced in comparison with the steam reforming process and 1.3 % in comparison with the autothermal reforming process, while maintaining an adequate purity value (99.95%).

3.4. Economic and Sustainability Evaluation

Using the economic evaluation tool of Aspen Hysys it was possible to compare (although it's not precisely calculated since it wasn't possible to include raw materials costs and other factors) both the capital costs and operating costs of each process route. Likewise, using the energy analysis tool it was possible to obtain and compare the total utilities used after energetic integration. As shown in Table 13, in all parameters analyzed, the dry reforming process advantages are demonstrated, as it costs less to install and operate and uses less utilities. In Table 14 are shown the results for the aforementioned sustainability metrics and carbon emission target (after energetic integration) for each simulated process.

Table 13. Data obtained using Aspen Hysys' economic evaluation and energy analysis tools

	Steam Reforming	Autothermal Reforming	Dry Reforming
Total Capital Cost (USD)	6607190	7243870	2717870
Total Operating Cost (USD Year⁻¹)	2864260	7146700	977129
Total Utilities (MJ h⁻¹)	52910	53760	40020

Table 14. Results obtained for the sustainability metrics and carbon emission

	Steam Reforming	Autothermal Reforming	Dry Reforming
Material Intensity	0.890	0.436	0.534
Energy Intensity (kJ kg⁻¹)	174574.889	317202.905	7850.293
Accumulation Factor	0.720	1.615	2.50 10 ⁻⁵
Carbon Emissions (kg h⁻¹)	2957	3004	2236

According to the obtained data, the dry reforming wastes more energy and produces less accumulation of methanol while it wastes more raw material than the steam reforming and less than the autothermal reforming.

It was also possible to compare, via Hysys energy analysis tool, the carbon emission of each process after energetic integration, also included in Table 13. The dry

reforming process emits less carbon than both steam reforming and autothermal reforming process, thus being less harmful to the environment.

4. Conclusions

It was possible to simulate the steam reforming, autothermal reforming and dry reforming processes using the software Aspen Hysys. It was demonstrated the advantages of using dry reforming as a substitute to steam reforming based processes to produce methanol is shown as a better alternative, in terms of raw methanol molar flow and fraction. It was also shown as more attractive in terms of energetic sustainability, less expensive to install and operate and more environmentally friendly as it emits less CO₂, while it wastes more raw material.

5. Future Works

Going forward, the processes can be simulated using a more detailed model that takes into account both thermodynamics and kinetics parameters (with prior kinetic studies to determine them) and an algorithm to obtain the best process conditions to optimize both economic, energetic and environmental parameters. It's also possible to simulate the process with a dynamic model, taking accumulation into account.

The processes simulated in this work can also be further compared with one-step processes of production of methanol from biogas, such as Direct Methane to Methanol (DMTM).

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